

# CSCI 6360

## Project 3

April 27, 2026

### 1 Random Forest Performance on Dataset 2

The Random Forest model was trained and evaluated on both in-sample and out-of-sample data. Hyperparameter tuning was performed using cross-validation on the training set to determine the optimal parameters. The tuning process considered the following parameters:

```
params = {  
    'n_estimators': [200, 300, 400, 500, 700],  
    'max_depth': [10, 20, 30, None],  
    'min_samples_split': [2, 5, 10],  
    'min_samples_leaf': [1, 2, 4],  
    'max_features': ['sqrt', 'log2', 0.5],  
    'bootstrap': [True, False]  
}
```

### 2 In-Sample Quality of Fit Metrics

Table 1: In-sample performance metrics of the Random Forest model.

| Metric | Value                 |
|--------|-----------------------|
| $R^2$  | 0.9549                |
| SST    | $9.78 \times 10^{11}$ |
| SSE    | $4.41 \times 10^{10}$ |
| MSE    | 441081.88             |
| RMSE   | 664.14                |
| MAE    | 374.50                |
| $m$    | 100000                |

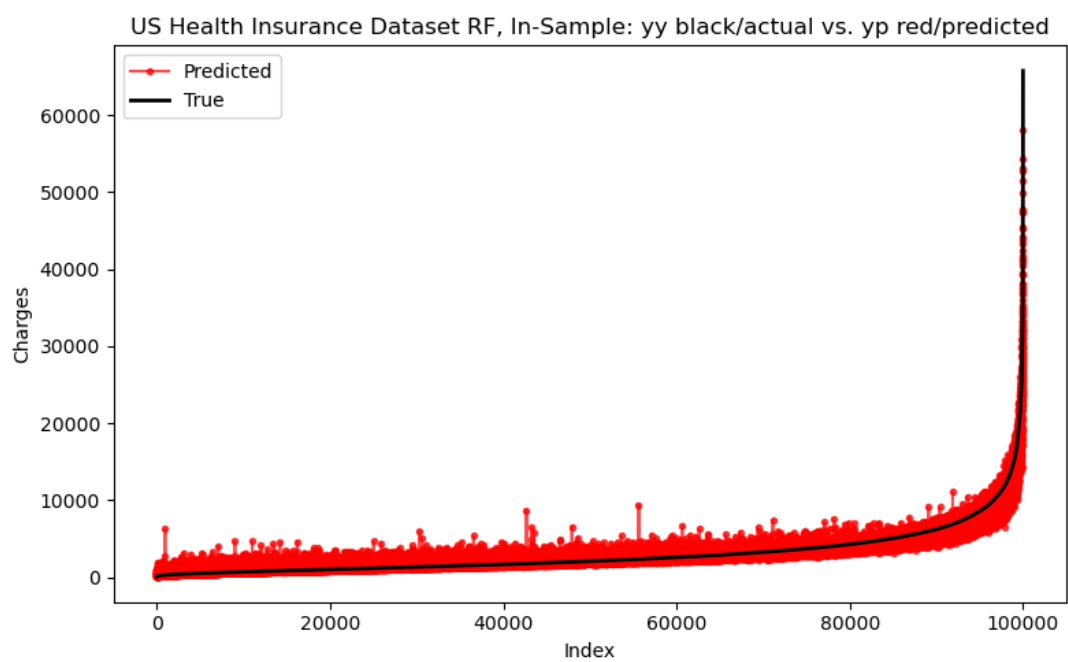


Figure 1: In-sample performance of the Random Forest model on Dataset 2.

### 3 Out-of-Sample Quality of Fit Metrics

Table 2: Out-of-sample performance metrics of the Random Forest model.

| Metric | Value                 |
|--------|-----------------------|
| $R^2$  | 0.6510                |
| SST    | $1.97 \times 10^{11}$ |
| SSE    | $6.87 \times 10^{10}$ |
| MSE    | 3434098.68            |
| RMSE   | 1853.13               |
| MAE    | 1025.39               |
| $m$    | 20000                 |

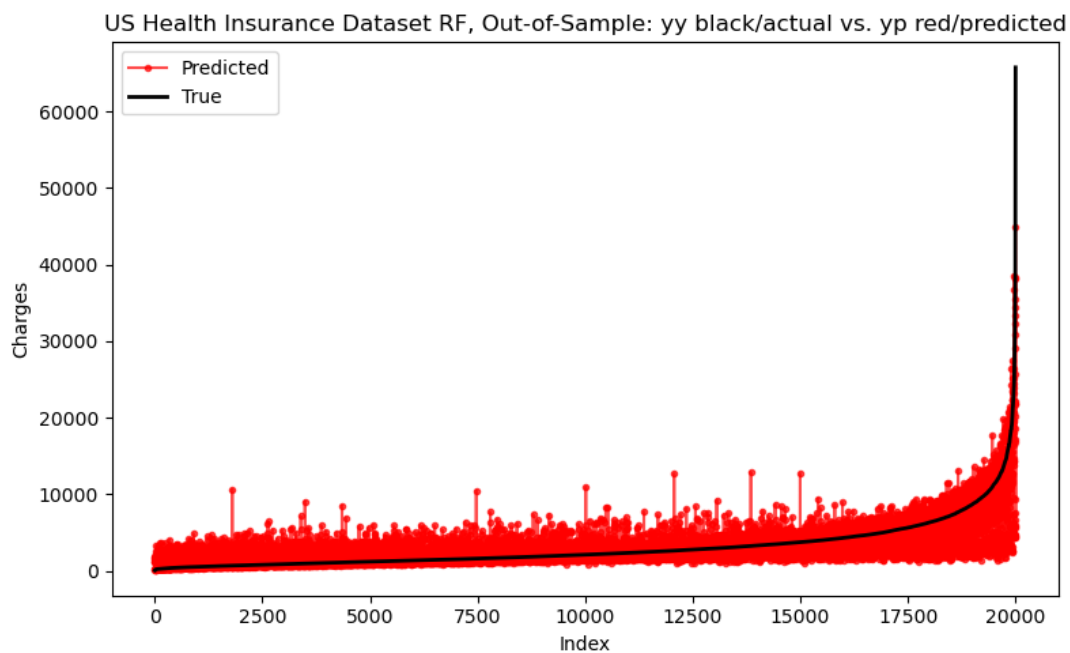


Figure 2: Out-of-sample performance of the Random Forest model on Dataset 2.

## 4 Fine-Tuned Quality of Fit Metrics

A total of 20 hyperparameter configurations were randomly sampled, and each was evaluated using 5-fold cross-validation. The following parameter settings were identified as optimal:

```
Best params: {'n_estimators': 300,
'min_samples_split': 5, 'min_samples_leaf': 4,
'max_features': 0.5, 'max_depth': None,
'bootstrap': False}
```

Table 3: Fine-tuned performance metrics of the Random Forest model.

| Metric | Value                 |
|--------|-----------------------|
| $R^2$  | 0.6542                |
| SST    | $1.97 \times 10^{11}$ |
| SSE    | $6.80 \times 10^{10}$ |
| MSE    | 3402240.89            |
| RMSE   | 1844.52               |
| MAE    | 999.62                |
| $m$    | 20000                 |

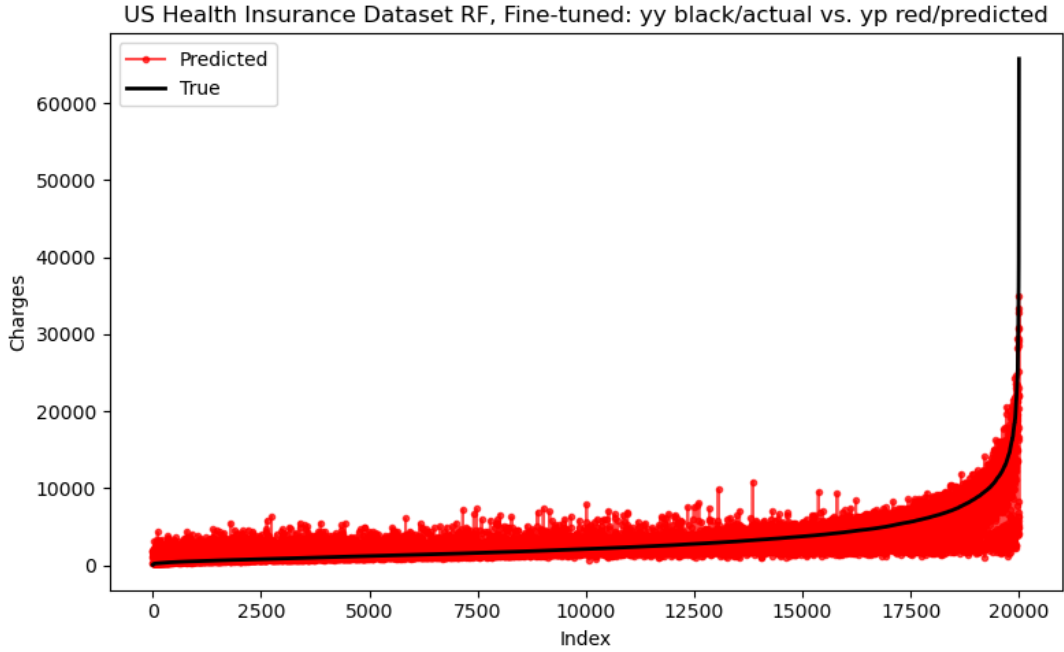


Figure 3: Fine-tuned performance of the Random Forest model on Dataset 2.

## 5 Feature Importance from the Best-Performing Model

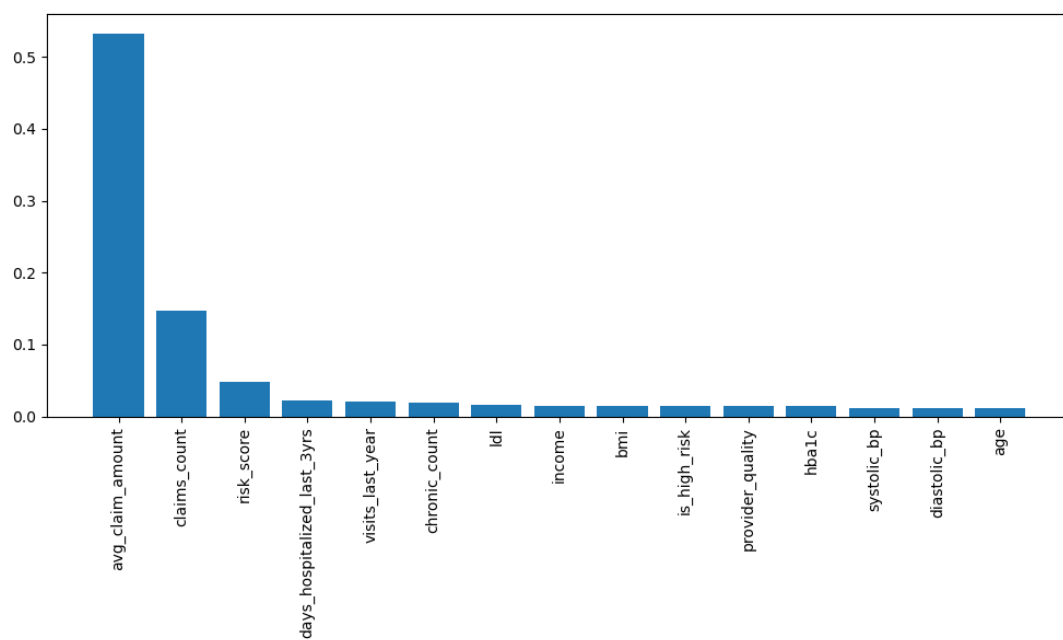


Figure 4: Top 15 Most Important Features.